Class XII Session 2025-26
Subject - Chemistry
Sample Question Paper - 10

Time Allowed: 30 minutes

General Instructions:-

Read the following instructions carefully.

1. There are 16 questions in this question paper with internal choice.
2. SECTION A consists of 16 multiple-choice questions carrying 1 mark each.
3. **All questions are compulsory.**
4. **Use of log tables and calculators is not allowed.**

Section A

1. Decreasing order of reactivity of hydrogen halide acids in the conversion of ROH → RX is: **[1]**

   a) HI > HBr > HCl > HF
   b) HF > HBr > HI > HCI
   c) HCI > HBr > HI > HF
   d) HF > HCI> HBr > HI

2. _____are joined together by phosphodiester linkage between 5′ and 3′ carbon atoms of the pentose sugar. **[1]**

   a) Nucleic acids
   b) Nucleotides
   c) Proteins
   d) Nucleosides

3. The conversion of an alkyl halide into alcohol by aqueous NaOH is classified as **[1]**

   a) a dehydrohalogenation reaction
   b) a dehydration reaction
   c) an addition reaction
   d) a substitution reaction

4. Pentan-2-one and Pentan-3-one can be distinguished by: **[1]**

   a) Tollens' test
   b) Sodium bicarbonate test
   c) Fehling's test
   d) Iodoform test

5. The unit of the rate of reaction is the same as that of the rate constant for a: **[1]**

   a) it cannot be same
   b) second order reaction
   c) zero order reaction
   d) first order reaction

6. Match the column and choose correct option **[1]**

Vant'Hoff factor	**Behaviour of compound**

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(a) i = 1	(i) Impossible
(b) i > 1	(ii) Association is the solution
(c) i < 1	(iii) Dissociation in the solution
(d) i = 0	(iv) No dissociation or association

a) (a) - (iv), (b) - (iv), (c) - (iii), (d) - (ii)
b) (a) - (iii), (b) - (iv), (c) - (ii), (d) - (i)
c) (a) - (iv), (b) - (iii), (c) - (i), (d) - (ii)
d) (a) - (iv), (b) - (iii), (c) - (ii), (d) - (i)

7. Racemisation occurs in **[1]**

a) Neither S_N1 nor S_N2 reactions
b) S_N2 reaction
c) S_N1 reaction
d) S_N2 reaction as well as S_N1 reaction

8. The catalyst used in Deacon's process is: **[1]**

a) Cu_2Cl_2
b) $CuCl_2$
c) $FeCl_3$
d) $FeCl_2$

9. The half-life for a zero order reaction equals: **[1]**
where R is the initial concentration.

a) $\frac{2k}{R}$
b) $\frac{R}{2k}$
c) $\frac{1}{2}\frac{k}{R^2}$
d) $\frac{R^2}{2k}$

10. **[1]**

The structure of A and type of isomerism in the above reaction are respectively.

a) Prop-2-en-2-ol, geometrical isomerism
b) Prop–1–en–2–ol, metamerism
c) Prop-1-en-2-ol, tautomerism
d) Prop-1-en-1-ol, tautomerism

11. The reaction $C_2H_5ONa + C_2H_5I \rightarrow C_2H_5OC_2H_5 + NaI$ is known as: **[1]**

a) Williamson's synthesis
b) Grignard's synthesis
c) Kolbe's synthesis
d) Wurtz's synthesis

12. Three compounds are given below: **[1]**

$\underset{I}{(C_2H_5)_3N}$, $\underset{II}{C_2H_5NH_2}$, $\underset{III}{(C_2H_5)_2NH}$

Identify the correct decreasing order of their basic strength in gas phase:

a) III > I > II
b) III > II > I
c) II > III > I
d) I > III > II

13. **Assertion (A):** Fructose is a reducing sugar. **[1]**
Reason (R): Fructose does not reduce Fehling solution and Tollen's reagent.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

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C) A is true but R is false. d) A is false but R is true. **[1]**

14. **Assertion (A):** Acetaldehyde undergoes aldol condensation with dilute NaOH.

Reason (R): Aldehydes which do not contain α-hydrogen undergo aldol condensation.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

15. **Assertion (A):** 2-Chloro-3-methylbutane on treatment with alcoholic potash gives 2-methylbut-2-ene as major product. **[1]**

Reason (R): The reaction occurs according to Saytzeff rule.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

16. **Assertion (A):** $(CH_3)_3C\text{-}O\text{-}CH_3$ gives $(CH_3)_3C\text{-}I$ and CH_3OH on treatment with HI. **[1]**

Reason (R): The reaction occurs by S_N1 mechanism.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

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$$CH_3 - C \equiv CH \xrightarrow[1\%HgSO_4]{40\%H_2SO_4} A \xrightarrow{\text{Isomerisation}} CH_3 - \underset{\underset{O}{\|}}{C} - CH_3$$

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